WORKED SOLUTION

STEP III 2008, Question 12

Probability and Statistics • Moment Generating Functions

STEP III 2008, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.24, *The Central Limit Theorem*: https://beyondstep.keithmcnulty.org/ch12/clt.html

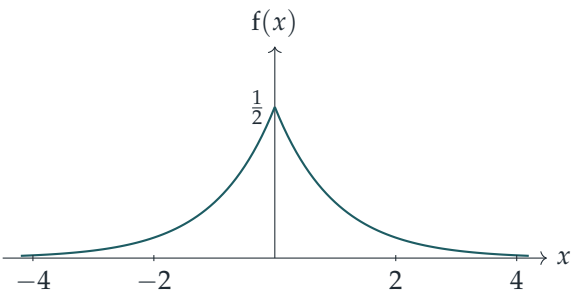
The key idea

The absolute value in the density splits every integral into two halves, $x < 0$ and $x > 0$, each of which is a plain exponential integral. After that, the whole question runs on two facts about moment generating functions: the m.g.f. of a sum of independent variables is the product of their m.g.f.s, and $M_{aY}(\theta) = M_Y(a\theta)$. Taking logarithms turns the nth power into n times something small, which is exactly what a Maclaurin series handles. This is the central limit theorem, seen in a case where you can do every step by hand.

Solution

The sketch

The density is even, has its maximum value $\frac{1}{2}$ at $x = 0$, where there is a sharp corner (the left and right derivatives there are $+\frac{1}{2}$ and $-\frac{1}{2}$), and decays exponentially to 0 on both sides.



The moment generating function

By definition $M_X(\theta) = E(e^{\theta X})$. Splitting at 0,

$$\begin{aligned} M_X(\theta) &= \tfrac{1}{2}\int_{-\infty}^{0} e^{(1+\theta)x}\,dx + \tfrac{1}{2}\int_{0}^{\infty} e^{-(1-\theta)x}\,dx \\ &= \frac{1}{2(1+\theta)} + \frac{1}{2(1-\theta)} = \frac{(1-\theta)+(1+\theta)}{2(1-\theta^2)} = \frac{1}{1-\theta^2}. \end{aligned}$$

The first integral converges only if $1+\theta > 0$ and the second only if $1-\theta > 0$, so this holds for $-1 < \theta < 1$, which is all we need near $\theta = 0$. This is the required result.

For the variance, expand as a geometric series for $|\theta| < 1$:

$$M_X(\theta) = 1 + \theta^2 + \theta^4 + \cdots.$$

Comparing with $M_X(\theta) = 1 + E(X)\,\theta + \frac{1}{2}E(X^2)\,\theta^2 + \cdots$, we get $E(X) = 0$ and $\frac{1}{2}E(X^2) = 1$. Hence

$$\mathrm{Var}(X) = E(X^2) - \big(E(X)\big)^2 = 2.$$

(Differentiating twice and setting $\theta = 0$ gives the same, but the series is quicker.)

The frog: m.g.f. of $Y/\sqrt{2n}$

Since the jumps $X_1, \ldots, X_n$ are independent, $e^{\theta X_1}, \ldots, e^{\theta X_n}$ are independent too, so the expectation of their product is the product of the expectations:

$$M_Y(\theta) = E\big(e^{\theta X_1}\cdots e^{\theta X_n}\big) = \prod_{i=1}^{n} M_{X_i}(\theta) = (1-\theta^2)^{-n}.$$

Writing $Z = Y/\sqrt{2n}$, we have $E(e^{\theta Z}) = E\big(e^{(\theta/\sqrt{2n})Y}\big) = M_Y\big(\theta/\sqrt{2n}\big)$, so

$$M_Z(\theta) = \left(1 - \frac{\theta^2}{2n}\right)^{-n}, \qquad |\theta| < \sqrt{2n}.$$

The scaling by $\sqrt{2n}$ is natural: $\mathrm{Var}(Y) = 2n$, so Z has mean 0 and variance 1.

The limit

Fix θ. Once $n > \frac{1}{2}\theta^2$ we have $0 \le \theta^2/(2n) < 1$, so we may take logarithms and use $\ln(1-u) = -u - \frac{1}{2}u^2 - \frac{1}{3}u^3 - \cdots$ for $|u| < 1$:

$$\begin{aligned} \ln M_Z(\theta) &= -n\ln\left(1 - \frac{\theta^2}{2n}\right) = n\left(\frac{\theta^2}{2n} + \frac{\theta^4}{8n^2} + \frac{\theta^6}{24n^3} + \cdots\right) \\ &= \frac{\theta^2}{2} + \frac{\theta^4}{8n} + \frac{\theta^6}{24n^2} + \cdots. \end{aligned}$$

Ask yourself How do we know the tail of that series really vanishes as $n \to \infty$?

With $u = \theta^2/(2n)$, everything after the first term is $n\left(\frac{1}{2}u^2 + \frac{1}{3}u^3 + \cdots\right) \le n \cdot \frac{1}{2}u^2(1 + u + u^2 + \cdots) = \dfrac{nu^2}{2(1-u)} = \dfrac{\theta^4}{8n(1-u)}$, which tends to 0 because $u \to 0$. So $\ln \mathrm{M}_Z(\theta) \to \frac{1}{2}\theta^2$, and since exp is continuous,

$$\mathrm{M}_{Y/\sqrt{2n}}(\theta) \to \exp\left(\tfrac{1}{2}\theta^2\right) \quad \text{as } n \to \infty,$$

as required.

The least number of jumps

Since the limit is the m.g.f. of the standard Normal, for large n we take $Z = Y/\sqrt{2n}$ to be approximately $\mathrm{N}(0,1)$. "Landing 25 cm or more from the start" means $|Y| \ge 25$, in either direction, so

$$\mathrm{P}(|Y| \ge 25) = \mathrm{P}\left(|Z| \ge \frac{25}{\sqrt{2n}}\right) \approx 2\left(1 - \Phi\left(\frac{25}{\sqrt{2n}}\right)\right).$$

This probability increases with n (the frog spreads out further), so we want the least n for which it reaches 5%. Since $\Phi(1.96) = 0.975$, we need

$$\frac{25}{\sqrt{2n}} \le 1.96 \iff 2n \ge \left(\frac{25}{1.96}\right)^2 \approx 162.7 \iff n \ge 81.3.$$

So the least number of jumps is $n = 82$. (With $n = 81$ the approximate probability is about 4.95%; with $n = 82$ it is about 5.09%.)

Checks

- $\mathrm{M}_X(0) = 1$, as every m.g.f. must, and M_X is even, matching the symmetry of the sketch.
- Numerical quadrature of $\int x^2 \mathrm{f}(x)\,\mathrm{d}x$ gives 2.0000, and of $\int \mathrm{e}^{0.3x}\mathrm{f}(x)\,\mathrm{d}x$ gives $1.0989 = 1/0.91$.
- With $\theta = 1$, $(1 - \frac{1}{2n})^{-n}$ is 1.6702 for $n = 10$ and 1.6508 for $n = 100$, closing in on $\mathrm{e}^{1/2} = 1.6487$.
- A simulation of 10^5 frogs making 82 Laplace jumps each had about 5.1% of them finish at least 25 cm from the start.

Common slips

- Using the one-tailed value 1.645: the frog can end up 25 cm away on either side, so both tails count.

- Writing $\mathrm{M}_{Y/\sqrt{2n}}(\theta) = \mathrm{M}_Y(\theta)/\sqrt{2n}$. Scaling a variable scales the *argument* of its m.g.f.
- Giving $n = 81$ by rounding 81.3 to the nearest integer: at $n = 81$ the chance is just under 5%.

Beyond STEP

What you have proved is the central limit theorem for Laplace steps, and the general proof follows the same route: expand $\ln \mathrm{M}(\theta/\sqrt{n})$ to second order and show the rest vanishes. In a first-year course you will meet the "continuity theorem", which justifies passing from convergence of m.g.f.s (or characteristic functions, which always exist) to convergence of probabilities, the step we took on trust here. Grimmett and Welsh, *Probability: An Introduction*, covers both.

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